

# 1 Divisibility

[Divides] Let  $a, b \in \mathbb{Z}$ . We say that  $a$  divides  $b$ , written  $a \mid b$ , if there exists  $k \in \mathbb{Z}$  such that  $b = ka$ . If no such  $k$  exists we write  $a \nmid b$ .

$3 \mid 12$  since  $12 = 4 \cdot 3$ , but  $5 \nmid 12$ . Every integer divides 0, and 1 divides every integer.

If  $a \mid b$  and  $b \mid c$  then  $a \mid c$ .

Write  $b = ka$  and  $c = \ell b$  for some  $k, \ell \in \mathbb{Z}$ . Then  $c = \ell b = \ell(ka) = (\ell k)a$ , and  $\ell k \in \mathbb{Z}$ , so  $a \mid c$ .

If  $a \mid b$  and  $a \mid c$  then  $a \mid (mb + nc)$  for all  $m, n \in \mathbb{Z}$ .

Write  $b = ka$  and  $c = \ell a$ . Then  $mb + nc = mka + n\ell a = (mk + n\ell)a$ .

Lemma 1 is the fact that makes the Euclidean algorithm work: the set of common divisors of  $a$  and  $b$  is unchanged when  $b$  is replaced by  $b - qa$ .

# 2 The division theorem

[Division with remainder] Let  $a \in \mathbb{Z}$  and  $b \in \mathbb{Z}_{>0}$ . There exist unique  $q, r \in \mathbb{Z}$  with

$$a = qb + r \quad \text{and} \quad 0 \leq r < b.$$

*Existence.* Consider the set  $S = \{a - kb : k \in \mathbb{Z}, a - kb \geq 0\}$ . The set is non-empty (take  $k = -|a|$ ), so by well-ordering it has a least element  $r = a - qb$ . If  $r \geq b$  then  $r - b = a - (q+1)b \in S$  is smaller than  $r$ , a contradiction. Hence  $0 \leq r < b$ .

*Uniqueness.* Suppose  $a = qb + r = q'b + r'$  with  $0 \leq r, r' < b$ . Then  $(q - q')b = r' - r$  and  $|r' - r| < b$ , so  $b \mid (r' - r)$  forces  $r' - r = 0$ , and therefore  $q = q'$ .

Every integer is either even or odd, and not both.

# 3 Greatest common divisors

For  $a, b \in \mathbb{Z}$  not both zero,  $\gcd(a, b)$  is the largest  $d \in \mathbb{Z}_{>0}$  with  $d \mid a$  and  $d \mid b$ . We also write  $\text{lcm}(a, b)$  for the least positive common multiple.

[Euclidean algorithm] Let  $a \geq 0$  and write  $a = qb + r$  as in Theorem 2. Then  $\gcd(a, b) = \gcd(b, r)$ .

By Lemma 1, every common divisor of  $a$  and  $b$  divides  $r = a - qb$ , and every common divisor of  $b$  and  $r$  divides  $a = qb + r$ . The two pairs therefore have the same common divisors, hence the same greatest one.

To compute  $\gcd(252, 198)$ :  $252 = 1 \cdot 198 + 54$ ,

$198 = 3 \cdot 54 + 36$ ,

$54 = 1 \cdot 36 + 18$ ,

$36 = 2 \cdot 18 + 0$ . The last non-zero remainder is 18, so  $\gcd(252, 198) = 18$ .

[Bézout] For  $a, b \in \mathbb{Z}$  not both zero there exist  $m, n \in \mathbb{Z}$  with  $ma + nb = \gcd(a, b)$ .

[Proof sketch] Run the Euclidean algorithm and substitute backwards; each remainder is an integer combination of the previous two, hence of  $a$  and  $b$ .

\*Exercises

[(i)]

1. Prove that  $\gcd(n, n+1) = 1$  for every  $n \in \mathbb{N}$ .

2. Find  $m, n \in \mathbb{Z}$  with  $252m + 198n = 18$ .

3. Show that  $\gcd(a, b) \cdot \text{lcm}(a, b) = ab$  for  $a, b \in \mathbb{Z}_{>0}$ .